RHYTHMGAME

APPROACH

- Get principle
- Find physical components
- Make circuit
- Write code
- Assign code to the hardware
- Make 3d model

- (Build irl)

COMPONENTS OF A RHYTHM GAME Software-ddd

- (Adjusting Margin: -+80ms)
- Player Input: buttonpressed 4 buttons
- **ONLY ONCE PER MARGIN: IS HIT?**
- 0:none
- 1: 1 button- 1 time randomly select
- … 4: 4 buttons- 4 time randomly select
- Music Player: plays tracks, switches tracks, dynamics of music, layers music/sfx
  - Indicates timeposition!==bpm
  - Wwise, fmod
- Metronome: Input bpbeat %4, bar%16, part depends STEADYCLOCK
  - Where are we?
  - m, make beatDurationSec=60/bpm (1/8 note), beatDurationMs= 60/bpm *1000, lastBeat equals 0
  - Import timepostion from the Music Player and compare it to beatDurationMs: if timepostion>=beatDurationMs
    - lastBeat++
    - Emit(„beat“, lastBeat)
    - Timeposition +=beatDurationMs
  - Error Margin (door)
    - activeBeat start/end = beatDurationMs +- margin

COMPONENTS OF A RHYTHM GAME Software-ddd 2

- Composer
  - Writes chart based off of MIDI/bpm and set/randomized
  - Next required input: button+beat ex: A3 or K4
  - Beatx4 randomized 0(none),1(L),2(R),3(both)
  - array
- Judge HIGHRESCLOCK
  - Composer: required input?
  - Player Input: Button right?
  - Metronome: Error Margin open? Beat correct?
  - Output: yes or no
  - Perfect 20 3, great 30 2, good 50 1, miss 100ms 0
  - Using 3 vectors NOT lists: each ´beatposition 3 (4)values:
  - (Beatposition), required keys, pressed keys, accuracy
- Referee (Manager) Start: Musicplayerstart, givebpmtometronome, givelvldatacomposer, EVERYONEGAMESTART
  - score
  - DisplayGame: Score according to judge
  - End: Music Player-MusicEnd,EVERYONEGAMESTOP,give final
  - Read chart of everyone – give visual feedback

DISPLAY HW

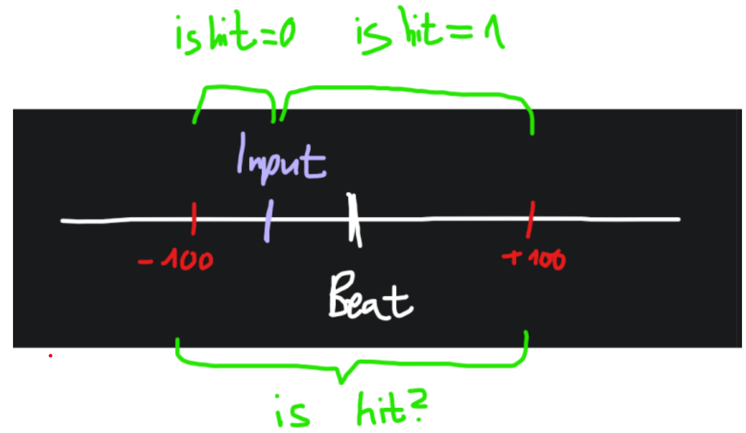
- LED Matrix
  - 64x32 RGB LED Matrix - 3mm pitch: https://www.adafruit.com/product/2279
    - Dimensions: 191mm x 96mm x 15mm / 7.5” x 3.8” x 0.6”
    - Panel weight with IDC cables and power cables: 189g
    - Compatible with M3 mounting screws
    - 5V regulated power input @ ~4A (with all LEDs on)
    - 2000 mcd LEDs on 3mm pitch
    - 1/16 scan rate
    - Indoor display, wide visibility
    - Displays are 'chainable
  - Adafruit RGB Matrix Bonnet for Raspberry Pi: https://www.adafruit.com/product/3211
  - Power
    - 5V 4A (4000mA) switching power supply - UL Listed: https://www.adafruit.com/product/1466
    - Female DC Power adapter - 2.1mm jack to screw terminal block: https://www.adafruit.com/product/368
  - Connection
    - Adafruit RGB Matrix Bonnet for Raspberry Pi: https://www.adafruit.com/product/3211
    - to calculate the power, multiply the width of all the chained matrices * 0.12 Amps : A 32 pixel wide matrix can end up drawing 32*0.12 = 3.85A

PROCESSOR HW

- Raspberry Pi:Model A, B, 2345 https://www.adafruit.com/categories/176
  - Depending on Code GB
  - C++ setup: https://www.raspberrypi.com/documentation/microcontrollers/c_sdk.html

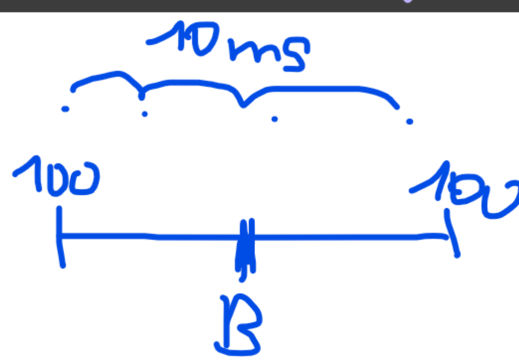
Raspberry Pi

Error margin



1: Generic

2: beat



Calc. w BPM?

?

last Beat Time 100

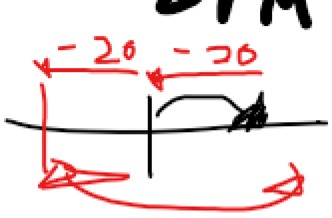
```
//margincheck
    if (leftpressed==1){
        auto emstart=std::chrono::steady_clock::now();
        milliseconds time{10};
            if((end - leftlastInputTime) >= margintime){

        //score margin
          if(lefttimediff<=perfect){
              cout<<"Lperfect"<<endl;
              accuracy.push_back(3);
          }else if(lefttimediff<=great){
              cout<<"Lgreat"<<endl;
              accuracy.push_back(2);
          }else if(lefttimediff<=good){
              cout<<"Lgood"<<endl;
              accuracy.push_back(1);
          }else if(lefttimediff<=miss){
              cout<<"Lmiss"<<endl;
              accuracy.push_back(0);
          }else{
              cout<<"doesnt count"<<endl;
              accuracy.push_back(0);
              keyinput.push_back(0);
          }
          leftbuttonact=1;
          leftpressed=0;
        }
```

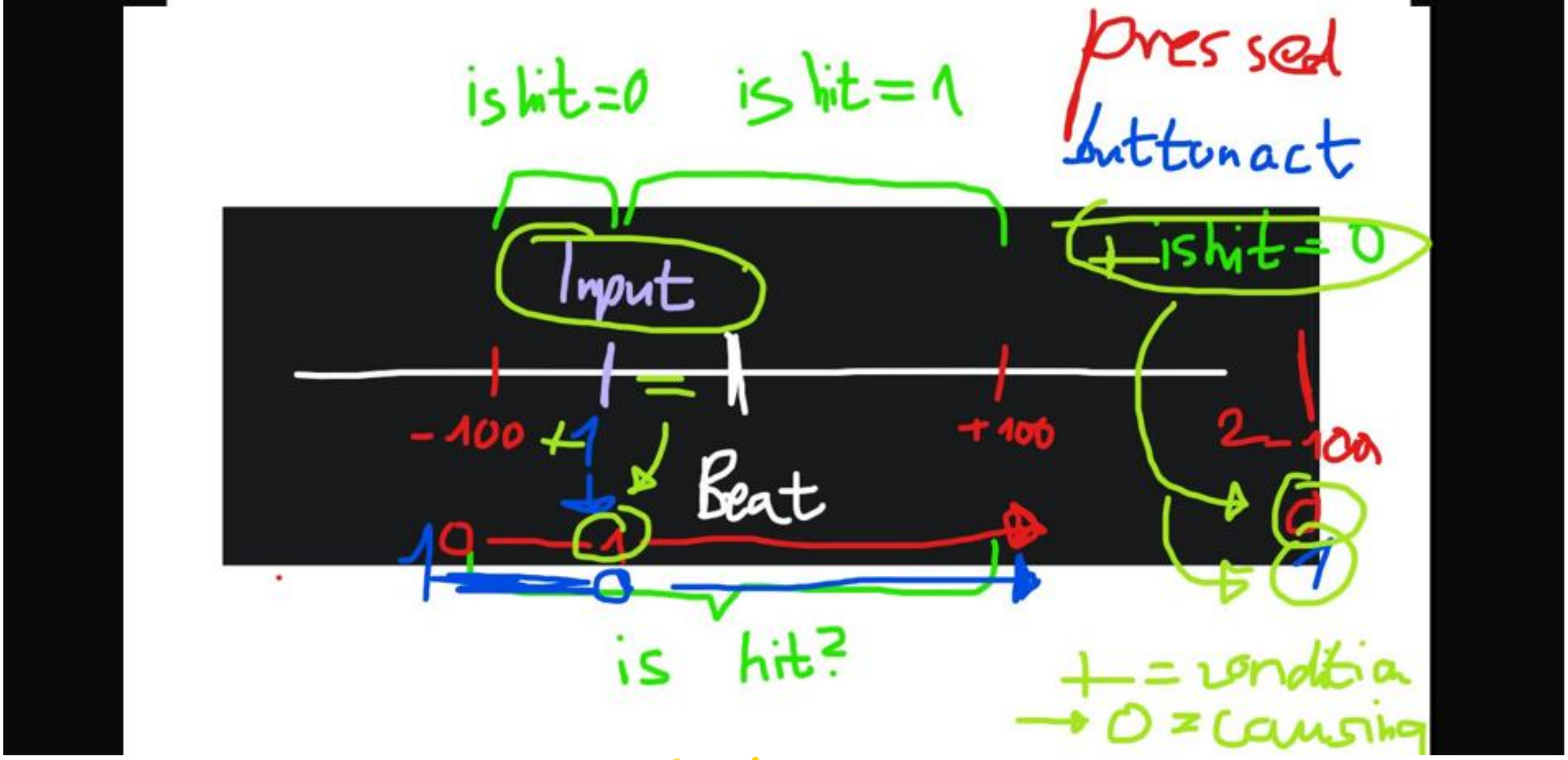
Signal

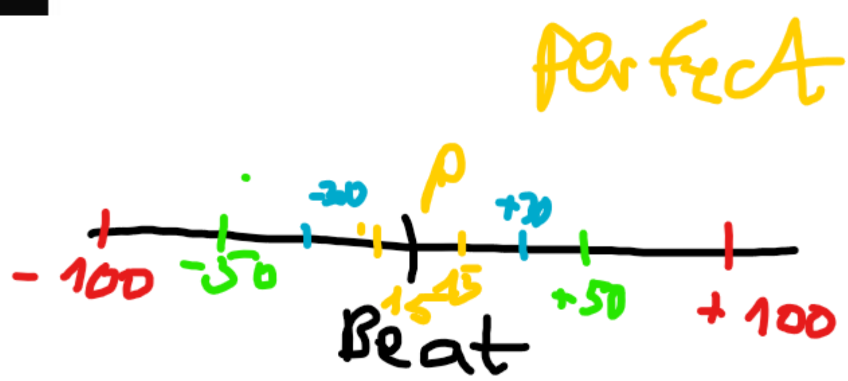
0 — No Input

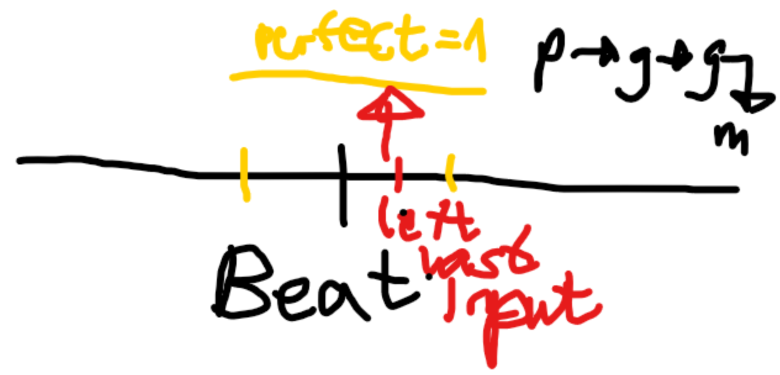
const ms

80 BPM









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S lB <= ba

Lmiss4

lB< ba

lB== ba

E

→ no action

First ideas

- Rhythm game: (Multiplayer) EXERCISE TOGETHER WHILE HAVING FUN!
-
- FNF type game on device with STOMPERS L and R
- Arcade gamed Model
-
- FNF game:
- Input a BPM & Difficulty
- LED matrix
-
- Microcontroller
-
- Gives SCORE
- Display https://www.adafruit.com/product/4311
-
- Animal slippers you have to sit down
- Senses the FNF
- sensors
-
-
- Stompers: light up lol
- LED stripes
- Hand controller:
- Senses the FNF
- Sensors
-
- ESP 32 MODEL: ??